

1. SDLC (Software Development Life Cycle)

Definition:

SDLC is a step-by-step process used by the software industry to design, develop, test, and deliver software. It ensures quality and systematic development.

Phases:

- Requirement Analysis
- Design
- Development (Coding)
- Testing
- Deployment & Maintenance

Key Points:

- Three pillars: **People, Process, Product**
- Ensures quality and reduces risk
- Different companies follow different SDLC models

Interview Importance:

Very commonly asked in interviews for QA, Developer, and Analyst roles.

Possible Interview Questions:

- What is SDLC?
 - Explain different phases of SDLC.
 - Why do we need SDLC?
 - What are the three P's in SDLC?
 - Give an example of SDLC in real life.
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2. Waterfall Model

Definition:

A traditional SDLC model where phases are executed sequentially. Each phase depends on the previous one.

Characteristics:

- Linear approach

- Heavy documentation
- No requirement changes allowed mid-process

Advantages:

- High-quality product due to clear documentation
- Less chance of bugs
- Initial investment is low
- Suitable for small projects with fixed requirements

Disadvantages:

- Requirement changes not allowed
- Defects in early phases propagate
- High cost for rework
- Testing only after coding

Interview Importance:

Frequently asked to check understanding of traditional models and their drawbacks.

Possible Interview Questions:

- What is the Waterfall Model?
 - Explain its advantages and disadvantages.
 - Why is Waterfall not suitable for modern projects?
 - How does Waterfall handle requirement changes?
 - Compare Waterfall with Agile or Spiral.
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3. Spiral Model

Definition:

An iterative SDLC model that allows requirement changes after each cycle. Each cycle includes all SDLC phases.

Characteristics:

- Iterative approach
- Suitable for product-based companies
- Allows requirement changes after each cycle

Advantages:

- Testing in every cycle
- Customer gets software early
- Requirement changes allowed after each cycle

Disadvantages:

- Requirement changes not allowed mid-cycle
- Each cycle resembles Waterfall
- No testing in design phase

Interview Importance:

Important for understanding iterative development and version control.

Possible Interview Questions:

- What is Spiral Model?
 - Why is it called iterative?
 - How does Spiral overcome Waterfall drawbacks?
 - When do you use Spiral Model?
 - Give real-world examples of Spiral Model.
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4. Prototype Model**Definition:**

A model where a sample (prototype) of the software is built first to get customer feedback before full development.

Key Points:

- Customer involvement is high
- Helps clarify requirements early
- No iterations like Spiral

Interview Importance:

Moderately asked, usually in context of requirement gathering.

Possible Interview Questions:

- What is Prototype Model?

- How is it different from Spiral Model?
 - Why do we use prototypes?
 - Give an example of prototype in real life.
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5. V-Model (Verification & Validation Model)

Definition:

An SDLC model where testing is integrated into every phase. Two sides: **Verification (documents)** and **Validation (software)**.

Key Features:

- Testing in every phase
- Static Testing (documents) vs Dynamic Testing (software)
- White Box vs Black Box Testing

Verification:

- Ensures we are building the right product
- Focus on documentation
- Techniques: Review, Walkthrough, Inspection

Validation:

- Ensures product works as intended
- Focus on software
- Techniques: Unit, Integration, System, UAT

Advantages:

- Early defect detection
- Less bugs in later stages

Disadvantages:

- High documentation
- High initial investment

Interview Importance:

Very important for QA roles and often asked in detail.

Possible Interview Questions:

- Explain V-Model with diagram.
 - Difference between Verification and Validation.
 - What is Static vs Dynamic Testing?
 - What is White Box vs Black Box Testing?
 - Why is V-Model better than Waterfall?
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6. Static vs Dynamic Testing

Static Testing:

- Testing documents (reviews, walkthroughs, inspections)
- Done before coding
- No execution of code

Dynamic Testing:

- Testing actual software
- Done after coding
- Includes Unit, Integration, System, UAT

Interview Questions:

- What is Static Testing?
 - Give examples of Static Testing techniques.
 - Difference between Static and Dynamic Testing.
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7. Verification vs Validation

Verification:

- Are we building the product right?
- Focus on process and documents
- Uses static testing techniques

Validation:

- Are we building the right product?
- Focus on actual software

- Uses dynamic testing techniques

Interview Questions:

- Difference between Verification and Validation.
 - Give examples of each.
 - Which comes first: Verification or Validation?
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8. White Box vs Black Box Testing**White Box Testing:**

- Tests internal logic/code
- Done by developers
- Examples: Unit Testing, Integration Testing

Black Box Testing:

- Tests functionality without knowing code
- Done by testers
- Examples: System Testing, UAT

Interview Questions:

- What is White Box Testing?
- What is Black Box Testing?
- Give examples of each.
- Why do testers use Black Box Testing?